

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CE/IT)

November 2012

IT205 - OPERATING SYSTEM

Date: 07.11.2012, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1** Consider the following set of processes in order P1, P2, P3 and P4 with the length of the CPU burst time given in milliseconds. Calculate average waiting time and turn-around time using shortest-remaining-time-first (pre-emptive) scheduling algorithm. Also draw Gantt Chart for SRTF scheduling algorithm. [08]

Process	Arrival Time	Burst Time
P1	0	8
P2	1	4
P3	2	9
P4	3	5

- Q - 2 (a)** A disk has 200 cylinders, numbered 0 to 199. The drive is currently serving a request at cylinder 53, and the queue of pending requests is: [08]

98, 183, 37, 122, 14, 124, 65, 67

Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for FCFS, SSTF and Elevator disk-scheduling algorithms?

- (b)** Explain the Banker's algorithm for single resource to avoid deadlock. [04]

OR

- (b)** What is deadlock? Which are the conditions for resource dealocks? [04]

- Q - 3 Answer the following questions. (Any Three)** [15]

1. Consider the page reference string: 7,0,1,2,0,3,0,4,2,3,0,3. How many page faults would occur for LRU page replacement algorithm? Initially main memory is empty.
2. A virtual address is 011001001100. MMU converts this address into 11-bit physical address. Page size is 256 bytes. Calculate number of pages and page frames.
3. Explain race condition and critical region with example.
4. Draw and explain 5-state process transition diagram.
5. Explain the terms: (1) Thrashing (2) Prepaging (3) Current Virtual Time

SECTION – II

Q - 4 (a) Write a shell script which takes one number from user and finds factorial of a given number. [04]

(b) What is page table? What is the significance of it? Mention fields of Page Table Structure. [04]

Q - 5 (a) Explain the operation of Direct Memory Access transfer. [05]

OR

(a) What is cryptography? Explain Secret-key and Public-key cryptography. [05]

(b) Explain the producer-consumer problem with respect to two processes. [04]

(c) Explain the three types of file structure. [04]

Q – 6 Attempt the following. (ANY TWO) [14]

1. Which are the algorithms that memory manager used to allocate free memory to a created process? Explain.
2. Discuss the Dining Philosophers problem related to processes synchronization.
3. Explain monolithic and microkernel system structures.

OR